# Embeddings and Vector Databases: Interview Q&A

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Five core questions on semantic search, each with a short interview answer, a simple example and a line to remember.

## 1. Why does keyword search fail for "server is out of space"?

**Answer:** Keyword search only matches the exact words typed. If the document is titled "Disk full", none of the words in "server is out of space" appear in it, so keyword search finds nothing, even though both mean the same thing.

### Example

| Search typed | Document title | Keyword search | Meaning search |
|---|---|---|---|
| server is out of space | Disk full | ❌ No shared words, so no match | ✅ Same meaning, so it matches |
| can't log in | Authentication failure | ❌ No match | ✅ Match |
| site is slow | High response latency | ❌ No match | ✅ Match |

**Interview line:** "Keyword search matches words, not meaning. Semantic search using embeddings finds documents that mean the same thing, even when they share no words."

**Remember:** keyword search = same words; semantic search = same meaning.

## 2. What is an embedding, in one sentence?

**Answer:** An embedding is a list of numbers (a vector) that represents the meaning of a piece of text, so that texts with similar meanings get similar numbers.

### Example

```
"disk is full"          → [0.12, -0.45, 0.88, ...]   ← similar numbers
"server out of space"   → [0.11, -0.43, 0.85, ...]   ← similar numbers
"user cannot log in"    → [-0.67, 0.29, 0.10, ...]  ← very different numbers
```

**Analogy:** think of every sentence as a point on a map. Sentences with similar meanings sit close together; unrelated ones sit far apart. A real map has 2 directions; an embedding usually has hundreds (for example 384, 768 or 1,536 numbers), which lets it capture many shades of meaning.

### Useful words

- **Vector:** another word for a list of numbers. An embedding is a vector.
- **Dimensions:** how many numbers are in the list.

- **Similarity:** how close two vectors are. Close to 1 means very similar meaning; close to 0 means unrelated.

**Interview line:** "An embedding converts text into a vector that captures its meaning, so similar meanings end up close together and can be found by similarity search."

**Remember:** embedding = meaning as numbers.

## 3. What's the difference between a chat model and an embedding model?

**Answer:** A chat model takes text and generates text: it answers, explains or summarises. An embedding model takes text and returns a vector of numbers that represents its meaning. Embedding models are used to search and compare; chat models are used to write the answer.

| | Chat model | Embedding model |
|---|---|---|
| Input | Text | Text |
| Output | Text (an answer) | Numbers (a vector) |
| Job | Generate, explain, summarise | Represent meaning for search and comparison |
| Typical use | Chatbots, summaries, drafting | Semantic search, finding similar documents, recommendations |
| Size and speed | Larger, slower | Smaller, faster |

**Who reads the numbers?** Nobody. Code (usually a vector database) compares the numbers to find the closest matches, then returns the original text. Humans and the chat model only ever see text.

**Analogy:** a maps app stores every restaurant's GPS coordinates (numbers). You never read the coordinates; the app compares them to find the nearest restaurants, then shows you their names.

**How they work together (RAG):** the embedding model finds the relevant text; the chat model reads that text and writes the answer.

**Interview line:** "A chat model generates text; an embedding model converts text into vectors for similarity search. In RAG, embeddings retrieve the right information and the chat model generates the answer from it."

**Remember:** embedding model = find; chat model = write.

## 4. Why do we cut documents into chunks?

**Answer:** If a whole document becomes one embedding, its meaning gets blurred into an average of everything in it, so search becomes imprecise. Splitting it into smaller chunks gives each part its own embedding, so search returns exactly the relevant section. Smaller pieces also mean fewer tokens are sent to the LLM, and they fit within its context window.

**Example:** a troubleshooting guide with three sections:

```
Guide: "Website is down"
├── Chunk 1: Symptoms   (error 502, timeouts)
├── Chunk 2: Diagnosis  (check service status, read error logs)
└── Chunk 3: Fix        (restart service, roll back last change)
```

A question like "how do I check why the site is down?" matches **Chunk 2** directly, instead of returning the whole guide.

### Three reasons

| Reason | Simple meaning |
|---|---|
| Precision | Smaller chunks have sharper meaning, so search is more accurate |
| Tokens | Send only the relevant part to the LLM, not the whole document |
| Context window | Big documents may not fit; small chunks always do |

### Chunk size is a trade-off (common follow-up question)

| Too big | Too small |
|---|---|
| Meaning is blurred; search is less precise | Pieces lack context ("restart it": restart what?) |
| Wastes tokens | The answer may be split across chunks |

Common approaches: split by headings or paragraphs, or by a fixed size with a small overlap between chunks, then test which retrieves best.

**Interview line:** "Chunking keeps each embedding focused on one topic, so search returns the relevant section and the LLM receives less, more useful text. Chunk size is a trade-off between precision and context, so it should be tested."

**Remember:** chunk so search finds the paragraph, not the whole book.

## 5. What three things does a vector database store for each chunk?

**Answer:** For each chunk, a vector database stores the original text, its embedding (the vector used for similarity search), and metadata such as the source document, section or date. The embedding is used to find matches, the text is what gets returned, and the metadata tells you where it came from, so you can cite sources and filter results.

| Stored | Example | Purpose |
|---|---|---|
| Text | "Restart the service with systemctl restart nginx" | What is returned to the user or the LLM |
| Embedding | [0.12, -0.45, 0.88, ...] | What is searched (by similarity) |
| Metadata | {"source": "website-down.md", "section": "Fix"} | Where it came from: citations and filtering |

Most vector databases also give each chunk a unique ID, like a primary key in a normal database.

### How a search works

1. The question is turned into an embedding.
2. The database finds the stored embeddings closest to it.
3. It returns those chunks' text and metadata.

### Normal database vs vector database

| | **Normal database (e.g. MySQL)** | **Vector database (e.g. ChromaDB, pgvector, Pinecone)** |
|---|---|---|
| Question style | Exact match: "where host = 'web01'" | Closest meaning: "most similar to this text" |
| Good at | Lookups, filters, totals | Similarity search |

**Interview line:** "A vector database stores each chunk's text, its embedding and metadata. It searches by comparing embeddings, returns the matching text, and uses the metadata for citations and filtering."

**Remember:** text to read, vector to search, metadata to cite.